

LESSON 301 (2.0 GROUND)

Lesson Objective

During this ground lesson, the student will be introduced to specific operations regarding weather, systems, and preflight.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Flight Operations Manual. (Version 6)
- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
 - Chapter 08 - Flight Instruments.
 - Chapter 12 - Weather Theory.
 - Chapter 13 - Aviation Weather Services.
 - Chapter 15 - Airspace.
 - Chapter 16 - Navigation.
- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 05 - Flight Instruments.
 - Chapter 09 - Navigation Systems.
- Warrior III Pilot's Information Manual.
 - Section 01 - General.
 - Section 07 - Systems.

Lesson Content

Operations Manual (FOM)

If this is a returning FIT student, quiz them on the FOM. If this is a new FIT student, go through the items below.

- Provide guidance to the student on how to access the FOM.
 - Fitaviation.com → students → forms and internal documents → login with TRACKS account.
 - Omit the "my" in the username. ex: [name@fit.edu](#) NOT [name@my.fit.edu](#) → manuals.
- Section 1 - Introduction.
 - Scheduling.

- Attendance.
- Flight Training Administration.
- Section 2 - Technical.
 - Weather Minimums.
 - Avoidance of Other Aircraft.
 - Ramp Operations.
 - Flight Operations.
 - Training Operations.
 - Testing and Certification.
 - Maintenance.
 - Emergency Operations.
- Section 3 - Restrictions and Limitations.
 - Dress Code
 - Alcohol and Drug Restrictions.
 - Alcohol and Drug Testing.
 - General Flight Restrictions.
 - Towered Airport Operations.
 - Non-Towered Airport Operations.
 - Solo Training Operations.
 - Time Building.
 - Instrument Training Operations.
 - Night Operations.
 - Glass Cockpit Operations.
 - Flight Time and Rest Limitations.
- Section 4 - College of Aeronautics Trainees.
 - Payment Method.
 - Duration/Completion of Flight Courses.
 - Grading.
 - Flight Course Failure/Withdrawal.
 - Not Completing a Flight Course.
 - Flight Fee Refund.
- Appendix 1 - SMS and Management Systems.
- Appendix 2 - Simulation Center Operations.

Pilot Qualifications

- Instrument Rating Requirements
 - Part 61:
 - [61.65](#)
 - At least a private pilot.
 - 50 hours of XC as PIC; 10 of these must be in airplanes.
 - 40 hours (15 with CFII) of actual or simulated instrument.

- 3 hours of training in preparation for check ride within 2 calendar months.
- 1 XC:
 - 250 NM.
 - Along airways or ATC-directed routing.
 - An instrument approach at each airport.
 - Three different kinds of approaches using NAV systems.
 - Filed IFR flight plan.
- Part 141:
 - [Appendix C of Part 141](#)
 - This information is found in the relevant course syllabus
 - At least a private pilot.
 - Stage 1 consists of basic attitude flying and intercepting and tracking.
 - Stage 2 consists of holds, approaches, and IFR cross countries.
 - 38 hours total time.
 - 35 hours instrument.
 - 14 hours of the instrument time is ATD/FTD.
- Privileges:
 - Fly in Class A - [91.135](#).
 - Special VFR at night (sunset to sunrise) - [91.157](#):
 - Airplane must have the required instrument equipment and the pilot must have an instrument rating.
 - A VFR pilot is privileged to fly below VFR minimums with a VFR clearance.
 - 1SM and COC.
 - File an IFR flight plan:
 - Clearance is only required before the pilot enters IMC in controlled airspace.
 - Can fly in IMC in class G but it is careless and reckless as per Administrator v Murphy.
 - Fly in instrument meteorological conditions (IMC)
 - [91.155](#)
 - IMC
 - conditions that require PIC to utilize sole reference to the instruments to maintain the desired route of flight.
 - IFR
 - conditions that the instrument rules apply (<3sm and/or <1,000ft ceilings).
- Instrument Currency
 - [61.57](#)
 - Airplane, FTD, ATD (or combination).
 - ATD has a basic replication of the flight deck.
 - Ex: Redbird, Mentor.
 - FTD has full replication of the flight deck, including flight control replication.
 - Ex: Seminole Sim.

- Within the 6 preceding calendar months (ATD or Airplane):
 - 6 approaches.
 - Holding procedures (minimum of 1 hold)
 - Intercepting and tracking courses.
- After the initial 6 months have passed, you have an additional 6 months to regain currency with a safety pilot, instructor, examiner, OR in the simulator (grace period).
- After the grace period ends, an IPC is needed:
 - Covers:
 - (i) Air traffic control clearances and procedures.
 - (ii) Flight by reference to instruments.
 - (iii) Navigation systems.
 - (iv) Instrument approach procedures.
 - (v) Emergency operations; and
 - (vi) Postflight procedures.
 - Conducted by a DPE, US Armed Forces (provided the IPC is for military personnel), company check pilot (Part 121, 125, and 135), CFII, or anyone authorized by the FAA.
 - Requirements are in the IRA ACS.
 - Guidance is located in AC 61-98D.
 - AATDs can be utilized for the majority of the IPC as specified in the Letter of Authorization issued for the device. However, the circling approach, the landing Task, and the multiengine airplane Tasks must be accomplished in an aircraft or FFS (Level B, C, or D). A BATD cannot be used for any part of the IPC.
- Safety pilots
 - 91.109:
 - A safety pilot must hold a minimum of a PPL in the same category, class, and type (if required) of the aircraft being flown.
 - A safety pilot is not required to have the endorsements required for the specific aircraft of use.
 - Ex: PPL ASEL can be a safety pilot for tailwheel airplanes if it's SEL even if not endorsed for tailwheel, but not for AMEL.
 - Cannot enter IMC unless the safety pilot is instrument rated and current.
 - Airplane (single-engine only) must have dual controls or is equipped with a single throw-over control wheel.
 - The Safety pilot must have adequate vision forward and side to side.
 - If not, a competent observer may substitute.
 - Safety Pilots CAN log PIC if they are ACTING as PIC. The person under the foggles can log PIC because SOLE MANIPULATOR of CONTROLS because the operation requires more than one pilot (61.51) (Simulated instrument conditions in VMC only).
 - Safety pilot cannot log XC time (Gebhart 2009)
 - A safety pilot is not a required crew member for the entire flight.

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- As per 61.65(d), only the pilot conducting the entire flight, including takeoff, landing, and en route flight, as a required flight crewmember may log cross-country flight time.

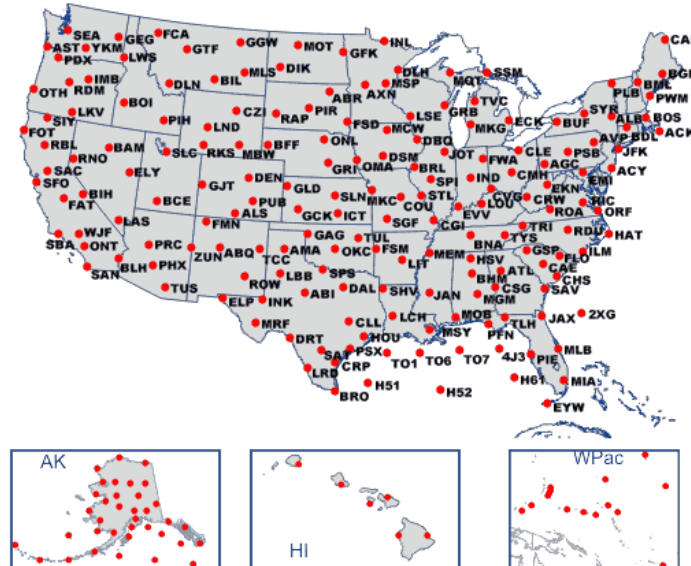
Weather Information

FAA-H-8083-28_Order_8083.28: Aviation Weather Handbook

Instructors, ensure your students are using all available resources when studying weather. Along with the PHAK, incorporate the Aviation Weather Handbook & AIM.

- Weather briefings:
 - Types of weather briefings:
 - Outlook.
 - Standard.
 - Abbreviated.
 - In-flight.
- METAR - Aviation routine weather report:
 - Observation of current surface weather.
 - Issued hourly unless significant changes have occurred (like wind direction changing by 45 degrees or more in less than 15 minutes, more in AC).
 - A special METAR (SPECI) can be issued between hourly reports.
 - Know how to decode them. Use the METAR decoding help sheet if needed.
- TAF - Terminal aerodrome forecast:
 - Established for a 5SM radius around an airport.
 - Valid for 24 up to 30 hours.
 - Updated 4 times a day.
 - Know how to decode them. Use the METAR decoding help sheet if needed.
- PIREPs - Pilot weather reports.
 - Weather report from a flight deck. It consists of message type, location, time, altitude/flight level, type of aircraft, and at least one other element to describe the report phenomena.
 - 2 types: routine (UA) and urgent (UUA):
 - Urgent includes:
 - Tornadoes.
 - Severe or extreme turbulence.
 - Hail.
 - Low-level wind shear within 2,000ft.
 - Required conditions to issue a PIREP:
 - Ceilings at or below 5,000 feet.
 - Visibility reported on the surface or aloft is 5 miles or less.
 - Thunderstorms and related phenomena.
 - Turbulence of moderate degree or greater.
 - Icing of light degree or greater.

- Wind shear.
- Braking action reports less than good.
- Volcanic eruption, ash clouds, and/or detection of sulfur gasses in the cabin.
- GFA Tool - Graphical forecasts for aviation.
 - Show the student where to find the GFA tool.
 - Go over the different items that the GFA tool can display:
 - Visibility, ceilings, cloud coverage, METARs, TAFs, precipitation, winds, thunderstorms, low-level wind shear, turbulence, and icing.
- Winds and temperatures aloft chart (Fb)
 - A summary of winds aloft forecast for the route/area as interpolated from forecast data for the local and/or the adjacent reporting locations for level through 12,000 feet.
 - Winds and temps at specific locations.
 - On a global scale, Issued 4 times a day (Every 6hrs) with forecasts out to 6, 12 & 24hrs.
 - The wind is in true knots.
 - The temperature is in Celsius.
 - If the wind is forecasted to be 100-199 knots, add 50 to the wind direction and subtract 100 from the speed.



- 731960
 - Wind is heading 230 at 119 knots.
 - The Temperature is -60 C.
- If the wind is at least 200 knots, the wind will be coded as 99 knots.
- 7799.
 - The wind is 270 and greater than 199 knots.
- Calm or less than 5 knots is 9900.
- NOTE:
 - No winds are forecasted within 1,500 feet AGL of station elevation.

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- No temperatures are forecasted within 2,500 feet AGL of station elevation.
 - Quiz the student on determining the atmosphere's stability using this chart!
- Foreflight Winds Aloft
 - Receives winds aloft forecast data from the Foreflight Global Wind Engine, providing winds and temperature data at any point on the planet up to FL540.
 - The Foreflight Global Wind Engine blends in observed data from the National Weather Service to accurately forecast turbulence, icing, winds & temperature across the world.
- Radar Imagery:
 - National radar:
 - Shows the intensity of precipitation.
 - Provides a single view of the entire country.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
 - Single site radar:
 - Composed of 140 radar sites across the U.S.
 - Each site shows base reflectivity, composite reflectivity, and total precipitation.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
 - ADS-B Radar imagery
 - ADS-B radar imagery has a lower resolution compared to internet radar due to bandwidth limitations. Similarly, regional radar imagery (within 250nm) has a higher resolution than the CONUS imagery.
- Significant weather prognostic chart:
 - Low-level significant weather-surface to FL240.
 - High level - FL250 to FL600.
 - 2 forms:
 - 12 and 24-hour forecast chart:
 - Issued 4 times a day.
 - The valid time is printed on the chart.
 - Shows forecasted weather to include non-convective turbulence, freezing levels, and IFR or MVFR weather.
 - Cover the difference between IFR, MVFR, and VFR.
 - IFR: less than 1000ft ceilings and less than 3SM visibility.
 - MVFR: 1000ft - 3000ft ceiling and 3-5SM visibility.
 - VFR: more than 3000ft ceiling and more than 5SM visibility.
 - 36 and 48-hour surface forecast chart:
 - Extension of the above chart.
 - Surface weather forecasts.
 - Issued 2 times a day.

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- Pressure patterns, fronts, and precipitation.
- Surface analysis chart:
 - Surface analysis charts – Chart depicts the distribution of several items, including sea level pressure, the position of the highs, lows, ridges, and troughs, the location and type of fronts, wind direction and speed, temperature, and dew point.
 - Analysis of current weather.
 - Computerized.
 - Transmitted every 3 hours.
 - Valid time is shown on the chart (observed data).
- AIRMETs, SIGMETs, and convective SIGMETs:
 - AIRMET (WA) – Advises of light to moderate weather that may be hazardous to aircraft (primarily light).
 - Airmet vs G-AIRMET
 - G-AIRMETs replaced the legacy textual airmets back in 2010.
 - G-AIRMET is a snapshot or a graphical depiction of a hazard at that given time using lat/long coordinates, whereas an AIRMET is a textual description using VORs to describe the affected area.
 - Before G-AIRMETs existed, pilots had to manually draw the affected area on a sectional chart after receiving a weather briefing.
 - G-AIRMETs are issued every 3 hours, whereas traditional AIRMETs are issued every 6 hours. AIRMETs are generated by the software using the initial, 3hr & 6hr forecast.
 - Sierra:
 - IFR conditions (ceilings less than 1000 ft, visibility less than 3sm).
 - Mountain obscurations.
 - Tango:
 - Light to moderate Turbulence.
 - Surface winds 30 knots or greater.
 - Low-level wind shear.
 - Zulu:
 - Icing.
 - Freezing levels.
 - An AIRMET will be issued if any of these conditions are expected to happen over an area of at least 3,000 square miles.
 - SIGMET (WS) – Advises of severe weather conditions that may be hazardous to all aircraft.
 - Unscheduled (comes out when needed).
 - Valid for 4 hours (6 for hurricanes).
 - Severe icing not associated with thunderstorms.
 - Severe or extreme turbulence or CAT not associated with thunderstorms.
 - Dust/sandstorms.
 - Volcanic ash.

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- Issued with an alphabetic identifier (no sierra or tango) → different zones throughout the USA have designated identifiers.
- Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on aircraft safety.
- Convective SIGMET (WST) – Hazardous convective weather affecting every flight.
 - Only issued for the contiguous United States.
 - Issued at 55 minutes past the hour.
 - Valid for 2 hours.
 - Severe thunderstorms with surface winds greater than 50 knots
 - Hail at the surface at least 3/4 inch in diameter
 - Affects 40% or more of a 3000 sq miles area.
 - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
 - Tornadoes
 - Numbered sequentially depending on the zone (West, Central, East).
 - If there is no convective weather, the chart will say “NONE”.
- Center Weather Advisory:
 - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.
 - CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
 - If conditions are expected to persist after the advisory’s valid period, a statement to that effect is included in the last line of the text. Additional CWAs will subsequently be issued as appropriate
 - CWA are issued by the NWS CWSU (Center Weather Service Unit). The 2 centers in Florida are Jacksonville & Miami.
- Convective outlook:
 - Consists of both a narrative and a graphic showing severe thunderstorm threats across the US.
 - Convective outlooks are split into 4 different charts:
 - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
 - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
 - Day 3: risk of severe weather for the next 24 hours. Issued daily.
 - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.
 - <https://www.weather.gov/jetstream/ahead>
- AWOS, ASOS, and ATIS:
 - Check the chart supplement to see what type of weather system is in use.
 - AWOS: Automated Weather Observing System.
 - Operated and controlled by the FAA.

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- Automated and reports the weather every minute.
- Types of AWOS:
 - AWOS-A: Altimeter setting.
 - AWOS-AV: Altimeter and visibility.
 - AWOS-I: Altimeter, wind, temperature, dew point, and density altitude.
 - AWOS-II: AWOS-I plus visibility.
 - AWOS-III: AWOS-II plus ceiling.
 - AWOS-III P: AWOS-III plus precipitation identification.
 - AWOS-III T: AWOS-III plus thunderstorm/lightning identification.
 - AWOS-III PT: AWOS-III plus precipitation and thunderstorm/lightning identification.
 - AWOS-IV: AWOS-III plus precipitation, type and accumulation, freezing rain, thunderstorm, and runway surface sensors.
- ASOS: Automated Surface Observing System.
 - Operated and controlled by the National Weather Service (NWS), Department of Defense (DoD), and sometimes by the FAA.
 - Used for the national weather system, not just for aviation purposes.
 - Reported information:
 - Wind speed and direction.
 - Visibility.
 - Sky condition.
 - Ceiling height and precipitation.
 - Barometric pressure.
 - Density altitude.
- ATIS: Automated Terminal Information Service.
 - Towered airports.
 - Requires a human to monitor and broadcast the ATIS.
 - ATIS information is obtained from the ASOS system.
 - Contains more than just weather (NOTAMs, remarks, etc.)
 - Published hourly at 55 minutes past the hour.
 - If the weather is rapidly changing, the tower can give an updated ATIS (like a SPECI METAR).
 - Reported information:
 - Phonetic alphabet letter.
 - Time of the report.
 - Wind speed and direction.
 - Visibility.
 - Cloud cover.
 - Temperature.
 - Dew point.

- Barometric pressure.
 - NOTAMs, remarks, other information.
- Thunderstorms
 - What you need for a thunderstorm:
 - Lifting action.
 - High moisture.
 - Unstable air.
 - Stages:
 - Developing stage:
 - Towering cumulus (rising air), little rain, occasional lightning.
 - Mature:
 - Fully formed, heavy rain, frequent lightning, strong winds. All hazards are at their worst; therefore, this stage is considered to be the most dangerous.
 - Dissipating:
 - Downdrafts, rainfall decreases, still producing strong winds and lightning.
 - Types:
 - Single cell: develop on warm, humid days. Severe, produce hail, and microbursts.
 - Usually associated with daytime convection.
 - Also referred to as air mass.
 - Multi-cell: develop in clusters, cover large areas, and can move in different directions.
 - Usually associated with large-scale weather systems such as low-pressure systems and fronts.
 - Also referred to as steady state.
 - Squall line: a narrow band of thunderstorms that form ahead of a cold front (moist, unstable air, lifting action). Severe and wide. Can move very quickly.
 - Supercell: Single long-lived thunderstorm. Large hail and significant tornadoes.
 - What to expect:
 - Lightning.
 - Turbulence.
 - Hail.
 - Tornadoes.
 - Wind shear.
- Clouds
 - High clouds [Cirrus (Ci), cirrocumulus (Cc), and cirrostratus (Cs)]
 - Cirrus
 - Composed of detached cirriform elements in the form of white, delicate filaments of white (or mostly white) patches, or of narrow bands
 - Have little effect on aircraft and contain no significant icing or turbulence.
 - Cirrocumulus

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- Appears as a thin white patch, sheet, or layer of cloud without shading, and is composed of very small elements in the form of grains, ripples, etc.
- Composed of highly supercooled water droplets, as well as small ice crystals, or a mixture of both; usually, the droplets are rapidly replaced by ice crystals. Pilots can expect some turbulence and icing.
- Cirrostratus
 - Appears as a whitish veil, usually fibrous (hairlike) but sometimes smooth, that may totally cover the sky, and that often produces halo phenomena, either partial or complete.
 - Composed primarily of ice crystals and contain little, if any, icing and no turbulence.
- Medium clouds [Altostratus (As), altostratus (As), and nimbostratus (Ns)]
 - Altostratus
 - White and/or gray in color, that occurs as a layer or patch with a waved aspect, the elements of which appear as laminae, rounded masses, rolls, etc. These elements usually are sharply outlined, but they may become partly fibrous or diffuse; they may or may not be merged.
 - At very low temperatures, however, ice crystals may form. Pilots flying through altostratus can expect some turbulence and small amounts of icing.
 - Altostratus
 - Gray or bluish (never white) sheet or layer of striated, fibrous, or uniform appearance.
 - Pilots can expect little or no turbulence, but light to moderate icing in the supercooled water regions.
 - Nimbostratus
 - A dark, gray wet-looking cloud characterized by more or less continuously falling precipitation (rain or snow) of light or moderate intensity (never heavy precipitation). It is not characterized by thunder, lightning or hail.
 - Produces very little turbulence but can pose a serious icing problem if temperatures are near or below freezing.
- Low clouds [Cumulus (Cu), towering cumulus (TCu), stratocumulus (Sc), stratus (St), and cumulonimbus (Cb)]
 - Cumulus
 - Individual, detached elements that are generally dense and possess sharp, non-fibrous outlines. These elements develop vertically, appearing as rising mounds, domes, or towers, the upper parts of which often resemble a cauliflower.
 - For cumulus with little vertical development, pilots can expect some turbulence and no significant icing. However, for towering cumulus (i.e., cumulus of moderate/strong development) pilots can expect very strong

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turbulence and some clear icing above the freezing level (where temperatures are negative). Towering cumulus is also referred to as the first stage of a thunderstorm.

■ Stratocumulus

- Predominantly stratiform, in the form of a gray and/or whitish layer or patch, which nearly always has dark parts and is non fibrous (except for virga).
- Composed of small water droplets, sometimes accompanied by larger droplets, soft hail, and (rarely) by snowflakes.
- Pilots can expect some turbulence and possible icing at subfreezing temperatures. Ceiling and visibility are usually better than with low stratus.

■ Stratus

- A gray layer with a fairly uniform base.
- Does not usually produce precipitation, but when it does occur, it is in the form of minute particles, such as drizzle, ice crystals, or snow grains.
- Produces little or no turbulence, but temperatures near or below freezing can create hazardous icing conditions. When stratus is associated with fog or precipitation, the combination can become troublesome for visual flying.

■ Cumulonimbus

- Exceptionally dense and vertically developed, occurring either as isolated clouds or as a line or wall of clouds with separated upper portions.
- Precipitation is often heavy and always of a showery nature.
- Contains large water drops, snowflakes, snow pellets, and sometimes hail.
- Contains nearly the entire spectrum of flying hazards, including extreme turbulence.

● Turbulence types:

- Light chop: slight rapid bumpiness.
- Light turbulence: a series of momentary changes in your altitude or attitude.
- Moderate chop: more intense than light chop.
- Moderate turbulence: changes in attitude and altitude, positive control though.
- Severe turbulence: large abrupt changes in altitude and attitude, tempo loss of control.
- Extreme turbulence: aircraft violently tossed around, impossible to control. Cause structural damage.
- CAT → severe turbulence in cloudless regions, usually at high altitudes with wind shear.

■ Frequencies:

- Occasional
- Intermittent
- Continuous

● Icing:

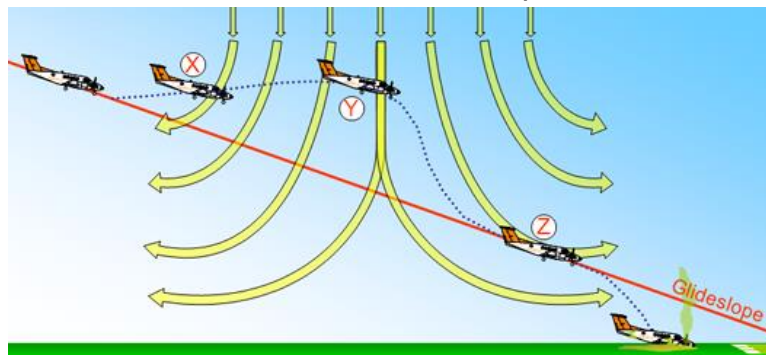
- Known icing vs known icing conditions:
 - Known icing: visible accretion (PIREPs).

- Known icing conditions: forecasts and reports (airmets, freezing level charts, winds and temps aloft, etc.).
- For an aircraft to be certified for flight into known or forecast icing conditions, it must demonstrate capabilities in accordance with 14 CFR Parts 23 & 25. Our aircraft are not certified.
- Structural:
 - Requires visible moisture and aircraft surface below freezing.
 - Clear → forms when large supercooled drop hits the aircraft and spreads and freezes slowly.
 - Cumulus clouds.
 - Clear in color.
 - Hits leading edge and freezes as it reaches trailing edge.
 - -10 to 0 Celsius.
 - Commonly found in cold fronts.
 - Rime → forms when small drops hit the aircraft and freeze rapidly.
 - White color.
 - Stratus clouds.
 - Freezes instantly.
 - -20 to -10 Celsius.
 - Commonly found in warm fronts.
 - Mix → mix of the two.
 - Found in thunderstorms where there is a mix of stratus and cumulus clouds.
 - It is also found in stationary or occluded fronts.
 - Frost → ice crystal deposits are formed when temp and dewpoint are below freezing.
 - Effects of icing:
 - Drag → increase.
 - Lift → decrease.
 - Weight → increase.
 - Stall speed → increase.
 - Icing intensities:
 - Trace.
 - Light.
 - Moderate.
 - Severe.
- Induction
 - Carb ice → can happen with temperatures as high as 100F/38C and a relative humidity of 80%.
 - Air in the carburetor is accelerated in the venturi, causing the temperature of the air to decrease. More susceptible to icing.

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- Throttle Ice → ice build-up in venturi due to the temperature of the air decreasing (faster air = colder air).
- Fuel vaporization ice → cooling effect when fuel mixes with air and vaporizes in the carburetor. Moisture condenses out of the air, and water starts freezing in the carburetor.
- Impact ice → As cold, moist air contacts anything solid, ice can start to form. This can happen on the air intake, filter, or carburetor.
- Turn carb heat on (small reduction in RPM, warm air is less dense than cold air, enriches the mixture).
- Instrument icing:
 - Icing tends to form on small exposed surfaces first:
 - Pitot tube and static ports.
 - Static wicks.
 - Antennas.
 - Stall warning tab.
 - OAT probe.
- Anti-Icing and Deicing Equipment:
 - Anti-Icing includes preventative measures.
 - Deicing equipment removes ice once it has occurred.
 - Some equipment serves both purposes.
 - Pneumatic Boots:
 - Air from the pneumatic pump, often from turbine bleed air.
 - Leading-edge inflatable boots blow out, breaking away the ice.
 - Some boots if exercised too often can cause the ice to conform to the boot.
 - Electric Boots:
 - Wiring heats up, attempting to melt ice.
 - Temperature is raised, preventing ice formation.
 - Thermal Anti-Ice:
 - Heat from the compressor section of a turbine engine.
 - Prevents ice from forming.
 - Alcohol Anti-Ice ("Weeping Wing"):
 - Prevents ice from forming.
 - Alcohol breaks up ice allowing it to be removed by aerodynamic forces.
 - "Thermawing":
 - Leading edges are lined with graphite foil laminate.
 - Material is electrically heated.
 - The forward section provides constant heat, and the aft section intermittently.
 - Propeller Anti-Ice:
 - These need to be anti-iced to prevent ice from being ingested by the engine.
 - Typical systems are alcohol-based or electrical.

- Windscreen Anti-Ice:
 - Typical systems are alcohol-based or electrical.
 - The electrical system can cause extreme compass deviation (40°).
- Pitot Heat:
 - This system melts accretions on the pitot tube/mast.
 - One may need an accurate airspeed indication to make an emergency landing if the airframe is icy.
 - There is no appreciable aerodynamic benefit from pitot heat.
- IFR conditions:
 - Ceiling less than 1,000 feet.
 - Visibility less than 3SM.
- Wind shear:
 - Dramatic shift in wind speed and/or direction.
 - 180 degrees or more.
 - Greater than 50 knots.
 - Low-level.
 - Result of temperature inversion.
 - Close isobars on charts indicate a strong pressure gradient = change in winds.
 - Winds flow across isobars at the surface due to friction and reduction of Coriolis force.
 - Winds flow parallel to the isobars at altitude (Winds aloft).
- Microburst
 - A strong downdraft that induces an outburst of damaging, divergent winds as high as 75 m/s on or near the ground over an area of 1-4 km.
 - Occurs in confined areas: less than one mile horizontally and within 1000 feet vertically.
 - Can span for 15 min.
 - Produce downdrafts of 6,000 feet per minute.
 - Can produce hazardous wind direction change of 45 degrees or more in a matter of seconds.
 - The plane will encounter an increase in headwind, then performance decreasing downdrafts, then a shear tailwind, and can result in terrain impact.



At point X, the aircraft enters the microburst zone where a headwind causes it to balloon above the normal glideslope. At the center of the microburst, point Y, there is a downdraft which causes the aircraft to sink. At point Z, the aircraft enters the most lethal zone where a sudden tailwind causes the aircraft to lose airspeed.

- Sea breeze and land breeze:
 - Land warms faster than water – air over land becomes warmer and less dense. It rises and is replaced by cooler, denser air flowing over the water.
 - This causes onshore wind called sea breeze.
 - At night – land cools faster than water, as does corresponding air. Warmer air over the water rises and is replaced by cooler, dense air from the land.
 - This causes offshore wind called land breeze.
- Stability of the atmosphere:
 - Stability of the atmosphere depends on its ability to resist vertical motion.
 - Rising air expands and cools due to the decrease in air pressure as altitude increases.
 - Opposite is true of descending air.
 - The combination of moisture (makes air lighter and enables air to rise) and temperature (high temperature leads to lifting action (air is expanding, less dense, lighter)) determine the stability of the air and the resulting weather.
- Inversions:
 - Surface-based temperature inversions occur on clear, cool nights when the air close to the ground is cooled by the lowering temperature of the ground.
 - The air within a few hundred feet of the surface becomes cooler than the air above it.
 - Frontal inversions occur when warm air spreads over a layer of cooler air.
- Fog:
 - Temperature of air near the ground is cooled to the air's dew point.
 - Types (reported when visibility is less than 5/8th of a mile):
 - Radiation fog – Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches the dew point. Does not form on cloudy nights because clouds trap heat. As the sun rises and the temperature increases, the fog burns off.
 - Advection fog – when a layer of warm moist air moves over cold land/water. 15-knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses.
 - Upslope fog – Occurs when moist, stable air is forced up sloping land features like a mountain range. Air is colder at higher elevations which cools air to dewpoint. Also requires wind for the formation and continued existence.
 - Steam fog – or sea smoke, forms when cold dry air moves over warm water. As water evaporates it turns into smoke.
- High and low-pressure systems:
 - In the northern hemisphere – the flow of air from high to low pressure is deflected to the right due to the Coriolis force.
 - High-pressure systems (outward - downward - clockwise).
 - Dry, stable, descending air.
 - Good weather.

- Low-pressure systems (inward - upward - counterclockwise)
 - Moist, unstable, rising air.
 - Bad weather.
- Fronts:
 - Air masses (body of air with equal or uniform properties (humidity, temperature, pressure, etc.)) form over regions and move the air over different areas which become subject to varying conditions of the land or water which modify the nature of the air mass.
 - Warm front:
 - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. The slope of the advancing front slides over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops, and condensation occurs from the humidity of the air.
 - Generally, cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary (rain evaporates when it gets close to the cooler ground).
 - Light to moderate steady precipitation.
 - Flight toward a warm front:
 - Visibility decreases.
 - Falling barometric pressure.
 - Weather overall deteriorates.
 - May be impossible to continue a VFR flight after flying through a warm front.
 - Cold front:
 - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
 - Moves at 25-30 mph, can move as fast as 60 mph.
 - Rapidly ascending air causes the temperature to decrease suddenly, forcing the creation of clouds.
 - It “lifts moist air and creates instability” - a recipe for thunderstorms.
 - Before the cold front passes, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
 - Showery precipitation.
 - After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in precipitation
 - Good visibility prevails and winds from the west and northwest
 - Flight toward a cold front:
 - Flight progresses to the oncoming cold front; the clouds show signs of vertical development with a broken layer.
 - Pilots should use sound judgment on frontal passage.

- Stationary Front:
 - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
- Occluded Front:
 - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.

Aircraft Systems Related to IFR Operations

- Required equipment for IFR - 91.205
 - You need BOTH VFR day and night equipment.
 - VFR Day (ATOMATOFLAMES):
 - Airspeed Indicator.
 - Tachometer (for each engine).
 - Oil Pressure gauge.
 - Manifold Pressure (for constant speed props, supercharged, turbocharged).
 - Altimeter.
 - Temperature Gauge (liquid-cooled engine).
 - Oil temperature gauge.
 - Fuel Quantity Gauge.
 - Landing Gear Position Lights (if retractable).
 - Anti-Collision Lights (aircraft certified after March 11th, 1996).
 - Magnetic direction indicator.
 - ELT.
 - Seatbelts.
 - VFR Night (FLAPS):
 - Fuses/resettable circuit breakers.
 - Landing Light (Electric) for hire.
 - Anti-Collision Lights.
 - Aircraft certified after August 11th, 1971.
 - Aircraft certified before that date STILL NEED anti-collision lights but the standards/requirements are different.
 - Position Lights.
 - Source of Electricity (Alternator/Battery).
 - IFR (GRABCARD):
 - Generator or alternator.
 - Rate of turn indicator.
 - Attitude indicator.
 - Ball (slip/skid indicator).
 - Clock (digital or analog with sweeping second hand)
 - Has to be installed in the aircraft. It can be a part of another system (ex:

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MFD's clock).

- Altimeter (sensitive altimeter).
- Radios/nav equipment appropriate for the flight.
- Directional gyro.
- DME (only required above FL240).
- VOR, TACAN, GPS, in Class B.

Aircraft Flight Instruments

- Pitot-static instruments (PHAK - Chapter 08):
 - The student should be able to draw and explain the basics of the pitot-static system.
 - Altimeter:
 - How does it work?
 - Casing is attached to the static port.
 - Aneroid Wafer is in casing sealed to 29.92in Hg (3 for redundancy).
 - Aneroid Wafer expands or contracts against static pressure in the casing, reading an increase or decrease in altitude.
 - In a climb, there is a decrease in static pressure, so the static pressure in the case exits, allowing the aneroid wafer to expand, showing an increase in altitude.
 - The opposite is true for a descent.
 - Gears and linkages connected to needles.
 - Pressure can be adjusted through the Kohlsman window → sensitive altimeter.
 - 1 in hg is equal to 1000 ft.
 - Types of altitudes:
 - Indicated altitude - Read off the altimeter.
 - Pressure altitude - Height above the standard datum plane.
 - Pressure altitude = Indicated altitude when Kohlsman set to 29.92" Hg
 - $PA = (Std. - current\ altimeter\ setting) \times 1000 + - field\ elevation.$
 - Density altitude - Pressure corrected for non-std temp.
 - $DA = PA + (120 \times (OAT\ ^\circ C - ISA\ temperature\ ^\circ C)).$
 - Higher DA = Worse performance.
 - Reduced Engine Output.
 - Reduced propeller efficiency.
 - Reduced Flight control effectiveness.
 - True altitude - Height above sea level (MSL).
 - True altitude = Indicated altitude when Kohlsman is set to the current pressure setting.
 - Absolute altitude - Height above the ground or terrain (AGL).

- Errors:
 - Flying from high pressure to low pressure (without adjusting the altimeter setting): Airplane will be lower than indicated.
 - This is because low pressure enters the altimeter case, so the aneroid wafers expand and indicate that the plane is higher. As a result, the pilot pitches down to return to the assigned altitude, making the plane lower than indicated.
 - Flying from hot to cold weather (without adjusting the altimeter setting): Airplane will be lower than indicated.
 - **REMEMBER: Look out below from hot to cold or high to low!**
- Vertical speed indicator:
 - How does it work?
 - Static port is connected to BOTH the CASING and DIAPHRAGM.
 - Calibrated leak (6-9 second lag) in the CASING causes static air to **leave** the casing **SLOWER** than the diaphragm leading to a pressure differential read as a vertical speed.
 - Measures rate of climb/descent.
 - Pressures are equalized when level, thus showing 0ft up or down.
 - Descent: diaphragm expands.
 - Climb: diaphragm contracts.
- Airspeed indicator:
 - How does it work?
 - Pitot tube connected to DIAPHRAGM.
 - Static port connected to CASING.
 - Diaphragm is like a balloon that expands as more air enters it. The expansion is read as an increase in airspeed. Resisting the expansion is the static air in the casing. The difference between static and dynamic is pressure differential.
 - Differential pressure (airspeed read) = static + ram (pitot).
 - Types of airspeeds (ICETGM):
 - Indicated airspeed - Read directly from the indicator.
 - Calibrated airspeed - Indicated airspeed corrected for installation/position error of pitot mast.
 - Equivalent airspeed - Calibrated airspeed corrected for compressible airflow error (>10,000 MSL and >250 kts).
 - True airspeed - Equivalent airspeed corrected for non-std temp and press.
 - Groundspeed - True corrected for wind.
 - Speed of movement across the ground – used for ETE).
 - Mach # - Ratio of true airspeed to the local speed of sound.
- Pitot-static blockages:

Problem	Airspeed	Altimeter	VSI
Blocked Pitot	Zero	Works	Works
Blocked Pitot and Drain Hole	High in climb Low in descent	Works	Works
Blocked Static	Low in climb High in descent	Frozen	Frozen
Alternate Static Air	Reads High	Reads high	Shows climb momentarily
Broken VSI	Reads High	Reads High	Reverses

- Using the alternate static source:
 - Prior to operation:
 - Close all vents in the cockpit and storm window.
 - Turn on the heater and defroster.
 - Instrument indications:
 - Airspeed indicator reads higher than normal.
 - Altimeter reads higher than normal.
 - VSI shows a momentary climb.
 - Lower pressure in the cabin due to leakage.
- Vacuum instruments:
 - Powered by an engine-driven dry-type vacuum pump → the lubricant is graphite. → WE DO NOT HAVE A VACUUM SYSTEM IN THE WARRIOR AVIDYNE.
 - Air enters the vacuum system through a filter, passes through the air-driven gyro instruments and the suction gauge, into the vacuum regulator, and on to the vacuum pump from which it is expelled through a short pipe.
 - There is a shear drive that protects the engine accessory drive from damage due to a seized pump.
 - The shear drive will disconnect the pump from the engine.
 - An annunciator light illuminates when suction is below normal operating pressure. (4.8-5.1 psi).
 - Vacuum system malfunctions:
 - Low suction due to:
 - Faulty vacuum regulator.
 - Dirty filter.
 - System leak.
 - Low engine RPM.

- No suction due:
 - Sheared pump drive.
 - Defective pump.
 - Defective gauge.
 - Collapsed line.
- What is a gyroscope?
 - A spinning disk that is able to maintain its position and orientation.
 - Operate based on two principles.
 - Rigidity in Space:
 - Ability of gyro to remain in a fixed position in the plane in which it is spinning.
 - Gimbals allow the aircraft to rotate freely in any direction and the gyro to remain fixed.
 - Precession:
 - Force felt 90 degrees in the same plane of rotation.
- Attitude indicator:
 - How does it work?
 - Based on rigidity in space.
 - Used to inform pilots of the orientation or attitude of the aircraft relative to earth.
 - Horizontally mounted gyro that rotates on the vertical axis with dual gimbals.
 - The front-facing horizon card is connected to the horizontal gimbal.
 - The airplane rotates about the gimbals.
 - Gyro rotates counterclockwise (18000-21000 RPM)
 - 4 air-ports covered by pendulous vanes use precession to stabilize the gyro (one port opens more than the other to reset the gyro upright).
 - This is used in case the gyro isn't upright.
 - Rotor tilts, but pendulous vanes remain vertical (gravity), exhaust port opens fully on one side, closes on other side, causing increased force from the open port. Precession causes force to act 90 degrees to that, erecting the gyro.
 - Our standby attitude indicator is electrically powered, meaning the gyro is spun using electrical power and not air.
 - Test its battery during the ground check.
 - Limitations:
 - Banking more than 110 degrees.
 - Pitching more than 70 degrees.
 - Errors:
 - Accelerating causes a false climb indication.
 - Decelerating causes a false descent indication.

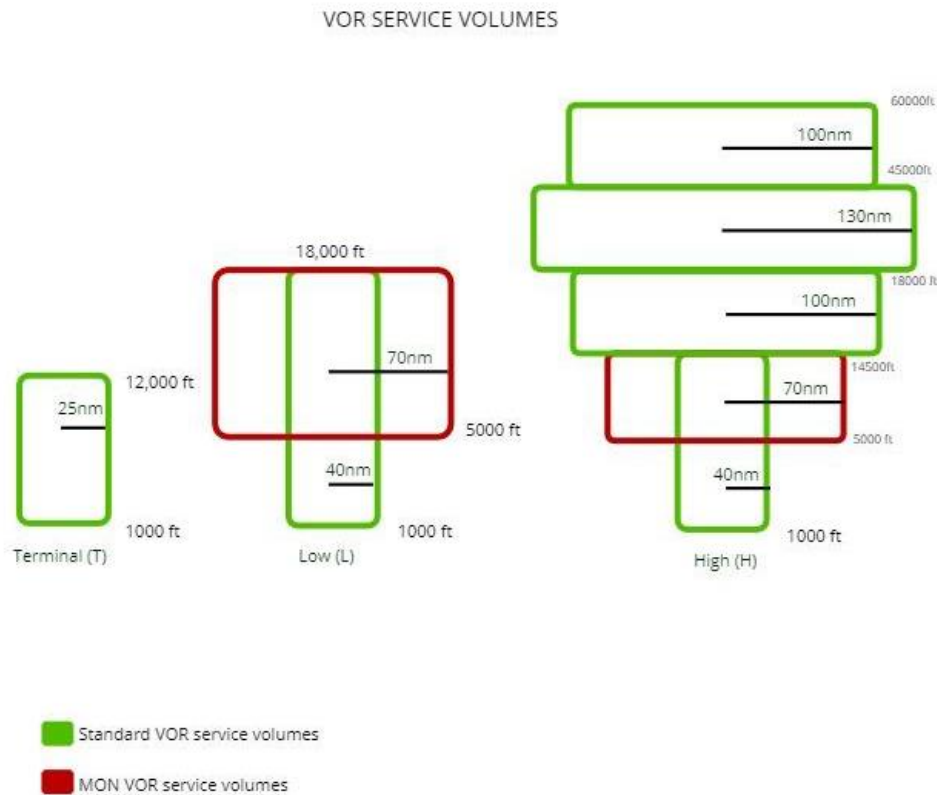
- Skid: shows turn in the opposite direction of skid.
- Directional Gyro (heading indicator):
 - How does it work?
 - Based on rigidity in space.
 - Vertically mounted gyro that rotates on the horizontal axis.
 - Spins away from the pilot.
 - Gyro speed: 10,000 - 18,000 RPM
 - Movement of the yaw gimbal transmitted to the instrument through gears.
 - Air nozzles: Two air jets will shoot air into the gyro if it precesses. They shoot air on one side to bring it back upright.
 - Limitations:
 - Exceeding 55 degrees of bank or pitch may cause the gyro to tumble (spin erratically) and become inaccurate.
 - Errors:
 - Error about 3 degrees every 15 minutes due to precession & friction.
 - Realign every 15 minutes with the mag compass.
- Turn-slip indicator/turn coordinator:
 - How does it work?
 - Based on precession.
 - Vertically mounted electrically driven gyro that rotates on the horizontal axis.
 - Rotates away from the pilot.
 - Cannot tumble because of the spring system.
 - Uses precession to determine the direction of movements (left/right).
 - Our turn coordinator is electrically powered.
 - Turn coordinator has a gyro that is canted back 30 degrees and a set of calibrated springs that allow it to sense rate-of-roll.
 - Rate of turn: the number of degrees of heading change per unit of time.
 - Rate of roll: measures the speed at which an airplane rotates around its longitudinal axis.
- Electrical system (Warrior III):
 - Battery:
 - 24 volts and 35 amp-hours.
 - Used for engine starting; powers the starter motor and gives initial charge to the alternator.
 - Used as a backup in case the alternator and standby alternator fail.
 - Alternator:
 - 28 volts and 70 amps.
 - Driven by the alternator belt connected to the engine.
 - The alternator belt spins the pulley, which spins the rotor (rotating electromagnet) inside the stator (copper windings).

- Standby alternator:
 - 28 volts and 20 amps (2500 RPM).
 - Activates if bus voltage drops to 26 volts.
- Voltage regulator:
 - Attempts to hold a steady electrical output (28V to 30.5).
- Overvoltage relay:
 - Takes alternator offline when the voltage regulator isn't properly regulating the voltage.
- Bus bar:
 - Electrical distribution system, like a power strip.
- Static wicks:
 - Static electricity is created throughout the fuselage (metallic) due to air friction.
 - Static wicks discharge static electricity off of the airplane.
 - Total of 9 static wicks.

Navigation Equipment

- Very high-frequency omnidirectional range (VOR):
 - 2 components: ground based VOR and your aircraft's receiver.
 - Ground VOR Emits two signals: variable and reference.
 - Variable Phase: VOR emits a signal per degree 30 times a second (kind of like a lighthouse).
 - Reference Phase: Sends out a 360 degrees pulse or broadcast of all of the radials.
 - The difference between the reference and variable phase is the phase shift and is calculated by your airborne VOR receiver to determine the radial you are on.
 - Frequencies: 108.0 - 117.95 Mhz - identified with a morse code identifier.
 - Errors:
 - Line of sight:
 - Obstacles, terrain, and slope of the earth can interfere.
 - Zone of ambiguity:
 - Occurs when perpendicular to a radial. The plane gets confused about whether it's TO or FROM.
 - Reverse sensing:
 - If the VOR is set to a TO indication, but you are actually coming FROM the station, you will have to turn opposite the needle.
 - Cone of confusion:
 - When the airplane is directly over the VOR, the VOR receiver will not be able to determine which radial the airplane is on.
 - If a VOR station is undergoing maintenance:
 - Either no morse code or test morse code (- -).

- If the VOR function is not working, but the DME function is:
 - Single code ID every 30 seconds.
- VOR types and service volumes:
 - VOR – The VOR by itself provides magnetic bearing information to and from the station.
 - VOR/DME – When DME (Distance Measuring Equipment) is also installed with the VOR.
 - VORTAC – When military tactical air navigations (TACAN) equipment is installed with a VOR
 - DME is always an integral part of a VORTAC.



MON INCLUDES STANDARD. MON IS AN EXPANSION

- 1000 ft guarantees that the aircraft will receive the signal. Under 1000 ft, it does not guarantee it, but it can still work.
- Check the chart supplement to see the type/service volume of a VOR.
- MON: Minimum Operational Network
 - The VOR MON program is optimizing the VOR network in the contiguous United States (CONUS) to provide a backup conventional service in the event of an unplanned Global Positioning System (GPS) outage.
 - In order to establish the VOR MON, new VOR Standard Service Volumes (SSVs) are being implemented to provide VOR signal coverage starting at 5,000' Above Ground Level

- (AGL).
- During an unplanned GPS outage, aircraft will be able to continue VOR-to-VOR navigation or proceed to a MON airport and conduct a VOR, Localizer LOC), or Instrument Landing System (ILS) approach.
- 306 VORs that do not meet the VOR MON criteria are planned for discontinuance.
- VOR checks:
 - VOR checkpoint (+/- 4°):
 - Located near a taxiway, ramp, or runup area. Follow instructions on the sign.
 - VOT (approved VOR test signal located at the airport) (+- 4 degrees):
 - Not all airports have one. Check chart supplement.
 - Tune VOR to VOT.
 - Set the course to 360 degrees, and the track indicator should be centered.
 - You should read FROM.
 - Next, set the course to 180.
 - And you should read TO.
 - Dual VOR check (+/- 4°):
 - Easiest check as long as you have 2 VOR receivers.
 - Tune both NAV radios to the same VOR facility.
 - Center the needle to a TO indication.
 - Airborne Check (+/- 6°):
 - VOR check in flight.
 - Select a VOR radial along the centerline of a VOR airway.
 - Select a ground point along the selected radial (preferably more than 20 nm) from the VOR facility.
 - Fly over the point and note the VOR bearing indicated by the receiver when over the point.
 - DATE, PLACE, ERROR, and SIGNATURE in the aircraft log (DEPS).
 - Every 30 days for IFR flight.
- Distance measuring equipment (DME):
 - Airborne DME unit transmits an interrogation signal that the ground facility receives and replies to the interrogation.
 - DME operates on UHF frequencies between 962 MHz and 1213 MHz.
 - The time passed is used by the airborne unit to calculate the slant range distance from the aircraft to the station $\text{Distance} = \text{Speed} \times \text{Time}$.
 - There is **not** a traditional DME receiver in the Avidyne. The G430 GPS substitutes for DME.
 - Errors:
 - Line of sight restrictions.
 - Maximum of 100 airplanes.
 - Slant Range Error:

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- For example, an aircraft directly above the DME station at 6,076 ft altitude would still show 1.0 NM on the DME readout
- The aircraft is technically a mile away, just a mile straight up
- Slant range error is most pronounced at high altitudes when close to the DME station.
- Global positioning system (GPS):
 - 3 components:
 - Space:
 - Satellite constellation.
 - Composed of a constellation of 32 satellites approx. 11,000 NM above the earth.
 - Must have 5 satellites in view at all times.
 - User:
 - Antennas and receivers on the aircraft.
 - Must be certified for the type of use (IFR, VFR, Backup).
 - Verifies the signal's integrity using Receiver Autonomous Integrity Monitoring (RAIM).
 - Control:
 - Ground-based monitoring system.
 - Master control station, 5 monitoring stations, and 3 ground antennas.
 - Ensures integrity of the satellite signal.
 - Corrects for distortion in satellite signal passing through the atmosphere.
 - How does it work?
 - At least 3 satellites are required for 2D position (latitude and longitude).
 - At least 4 satellites are required for 3D position (latitude, longitude, and altitude).
 - Time-lapse between transmission broadcast from atomic clocks on satellite to the time it was received by the aircraft transceiver is translated into distance. Using the intersection of spheres from multiple satellites, the aircraft's position is found.
 - Modes:
 - > 30 miles away:
 - Enroute Mode +/- 5NM (Non-WAAS) +/- 2NM (WAAS).
 - Switches to Terminal mode within 30 NM of the airport reference point.
 - < 30 miles away:
 - Terminal Mode +/- 1NM (Non-WAAS and WAAS).
 - Switches to Approach mode when 2 NM from the FAF.
 - 2 miles from FAF:
 - Approach Mode +/- 0.3NM (Non-WAAS and WAAS).
 - MAWP (Missed approach waypoint)
 - Suspend Mode +/- 1 NM

- GPS certification:
 - GPS may be used as a primary means of navigation if:
 - The receiver is certified for the type of use.
 - TSO C129 or C196: certifies GPS for non-WAAS.
 - TSO C145 and C146: certifies GPS for WAAS.
 - The database is current.
 - RAIM signals can be read.
 - WAAS (Wide Area Augmentation System) can substitute for RAIM.
- Receiver autonomous integrity monitoring (RAIM):
 - Function that monitors the integrity of the satellite signals.
 - RAIM check is required before the flight. Otherwise, backup equipment must be used that is not GPS based.
 - Fault Detection:
 - Requires a minimum of 5 satellites or 4 with baro-aiding altimeter.
 - Detects the presence of a positioning failure.
 - Fault Exclusion:
 - Requires 6 satellites or 5 with a baro-aiding altimeter.
 - Eliminates a corrupt satellite.
 - Upon detection, proper fault exclusion determines and excludes the source of the failure, thereby allowing GNSS navigation to continue without interruption.
 - Barometric aiding:
 - An integrity augmentation that allows a GPS system to use a non-satellite input source such as the aircraft pitot-static system to provide vertical reference.
 - If RAIM is not guaranteed, a signal loss could be due to:
 - Not enough satellites.
 - Antenna location on the aircraft.
 - Satellite malfunction.
- Wide area augmentation system (WAAS):
 - Improves the accuracy, integrity, and availability of GPS signals.
 - Allows for GPS approaches with vertical guidance (LPV or LNAV/VNAV).
 - No RAIM check is required, but you need to check WAAS NOTAMS.
 - How does it work?
 - WAAS Reference stations on the ground detect errors in GPS signals.
 - The errors from the reference stations are sent to WAAS master stations and corrected.
 - Corrected signals are sent to a ground uplink station.
 - The ground uplink station sends signals to geostationary satellites.
 - Geostationary Satellites send the corrected signal to the aircraft's receiver.
- Ground-based augmentation system (GBAS):

- Increases the accuracy of GPS and enhances the capabilities of the aircraft for approaches (GLS).
- GBAS works similar to WAAS but is used just in the vicinity of an airport.
- How does it work?
 - 4 Reference stations located around an airport receive signals from GPS satellites.
 - The reference stations find the error in the satellite transmissions and send the computed information to the ground-based Facility.
 - Ground-based Facility corrects the information and, through a VHF data broadcast, sends the augmented signal to the aircraft where it can now be used for approaches with minimums as low as 200 feet AGL.
- Instrument landing system (ILS):
 - An electronic system that provides horizontal and vertical guidance to a specific runway, used to execute a precision instrument approach procedure.
 - Components:
 - Localizer (AIM 1-1-9): Provides horizontal guidance along the centerline of the runway
 - Located on the extended centerline.
 - Guidance is given from 18 nm from the antenna up to 4,500' above antenna elevation.
 - First 10 nm → 35 degrees on each side of the centerline.
 - Remaining 8 nm → 10 degrees on each side of the centerline.
 - Provides course guidance between 108.1 and 111.95 MHz (odd tenths only). 90 and 150 Hz signals are sent by the antenna and used by the receiver to determine the position on the localizer.
 - Course width: 3-6 degrees (usually 5) and full-scale deflection: 2.5 degrees.
 - Glideslope (AIM 1-1-9): provides vertical guidance toward the runway touchdown point.
 - Frequencies: 329.3 to 335 MHz (automatically tuned with the localizer).
 - Vertical position is determined by the aircraft using the 90 and 150Hz signals sent by the glideslope antenna.
 - Equipment is housed in a building approx. 750-1250' down from the approach end of the runway and 400-600' to one side of the centerline.
 - Range: 10 NM.
 - Width: 1.4 degrees and full-scale deflection is 0.7 degrees.
 - Error: false glideslope.
 - Marker beacons: provide range information along the approach path (transmits at 75MHz)
 - Signal is directed upward, and when the aircraft is directly over it, an aural and visual indication appears inside the aircraft.
 - Outer marker (OM):

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- Blue and (- - -).
- Located 4-7 miles out and indicates the position at which the aircraft intercepts the glideslope (at the appropriate altitude).
- Middle marker (MM):
 - Amber and (- . -).
 - Located 3,500 ft from the landing threshold and indicates the point where the glideslope meets the decision height (usually 200ft above touchdown zone elevation).
- Inner marker (IM):
 - White and (. . .).
 - Located between the middle marker and landing threshold. It indicates the point where the glideslope meets the decision height on a CATII ILS.
- Back course marker (not part of the ILS system):
 - White and (. . .).
 - Indicates the final approach fix on a back-course approach.
- Approach lights: Assist in the transition from instrument to visual flight.
 - They extend:
 - 2,400 - 3,000ft for precision instrument runways.
 - 1,400 - 1,500ft for non-precision instrument runways.
 - Visual identification of the ALS must be instantaneous, so it's important to know the type:
 - ALSF, SSALR, MALSR, REL, MALSF, ODALS, and VASIs.
- Simplified directional facility (SDF):
 - Provides lateral guidance similar to a localizer but less accurate.
 - Width: 6 OR 12 degrees.
 - The antenna can be offset from the runway centerline but by no more than 3 degrees.
 - No vertical guidance.
- Localizer type directional aid (LDA):
 - More precise than SDF because the course is 3-6 degrees.
 - May provide a glideslope. If so, it is published as an LDA/Glideslope approach.
 - Not aligned with the runway. It can have straight-in minimums if the alignment is within 30 degrees from the runway centerline.
- Area navigation (RNAV):
 - RNAV is area navigation. It allows navigation on any paths without the need to overfly ground-based facilities.
 - Types:
 - GNSS (Global navigation satellite system) (ex: GPS).
 - VOR/DME RNAV.
 - DME/DME RNAV.
 - RNAV VNAV: Vertical navigation guidance.

- Baro-VNAV: an RNAV system that uses the barometric altitude to compute vertical guidance for the pilot.
- Required navigation performance (RNP):
 - What is it?
 - A statement of navigation equipment and service performance.
 - RNAV with navigation monitoring and alerting.
 - All RNAV approaches are RNP approaches.
 - RNP approach minima and equipment:
 - GLS DA minima using GBAS.
 - LP MDA or LPV DA minima require RNP achieved by WAAS.
 - LNAV /VNAV DA is achieved by VNAV-approved WAAS or baro-vnav systems.
 - LNAV MDA- achieved by a basic, unaugmented IFR-approved GPS.

Primary and Supporting Instruments / Control Performance Method

- Pitch + power = performance.
- Full panel:
 - Control instruments: tachometer and attitude indicator.
 - Performance instruments: airspeed indicator, altimeter, VSI, HSI, attitude indicator, and turn coordinator.
 - For example:
 - Straight and level flight at 90 knots (performance) = pitch (2.5 degrees) + power (2,300RPM).
- Partial panel:

Condition	Primary Pitch	Primary Bank	Supporting Pitch	Supporting Bank
S+L	Altimeter	DG	Attitude, VSI	Attitude, TC
Constant airspeed climb	Airspeed	DG	Attitude, VSI	Attitude, TC
Establishing level standard rate turn	Altimeter	Attitude	Attitude VSI	TC

Aircraft Avionics

- Electronic flight instruments (components of the PFD):
 - Air data computer (ADC):
 - Located on the back end of the PFD unit.
 - Provides digital airspeed, altitude, VS, and outside air temperature (OAT) information and computed winds aloft and true airspeed (TAS) to the PFD for display.
 - ADC is fed information from the pitot mast.
 - Ram and static lines are piped directly to the ADC.
 - The pressure transducers are in the ADC.

- The Outside Air Temperature Probes are used to sense and send OAT to the air data computer (ADC) for processing.
 - Located under the left wing.
 - Another one is located in the MFD and monitors system temperature too.
- The ADC gathers this information and uses it to compute pitot/static data and translate it to readable data on the PFD.
- If there is an ADC error, red X's will display where ADC instruments are.
 - Pitot static blockages DO NOT mean red X's.
- Attitude and heading reference system (AHRS):
 - Located in the back of PFD.
 - Provides attitude, heading, and turn rate data.
 - Receives information from the:
 - Magnetometer (or flux gate):
 - Senses the Earth's magnetic fields and electrically transmits that information to the HSI.
 - HSI combines a heading indicator, omni-bearing selector, and bearing pointer.
 - HSI does not precess as there is no gyroscope.
 - Located on the left wing to be as far away from magnetic disturbances.
 - Accelerometers:
 - Takes measurements from 3-axis solid-state inertial sensors similar to tuning forks (pitch, roll, yaw).
 - With 3 of these tuning forks mounted on three different axes (90 degrees from each other), any change in acceleration (plane) is detected as a change in voltage.
 - Each tuning fork vibrates at a specific frequency when the aircraft's attitude is changed. It changes the vibration frequency and, in result, changes the voltage that is then translated to the PFD as an attitude.
- PFD Limitations (PIM - section 09, page 9-64):
 - IFR flight is prohibited when the PFD or any standby instrument is inoperative.
 - IFR flight is prohibited upon aircraft total loss of essential engine parameter display.
 - The Avidyne FlightMax Entegra series Primary Flight Display Pilot's Guide must be available to the pilot during all flight operations.
 - If VLOC is displayed on the HSI and GPSS mode is engaged on the autopilot, the autopilot will track the active flight plan in the GPS corresponding to the selected VLOC.
 - GPSS mode must not be used on the final approach segment of a VLOC approach.
- Electronic flight instruments (components of the MFD):
 - Data acquisition unit (DAU):

- DAU receives information from pressure transducers and sensors throughout the engine (EIS - Engine Indicating System).
- MFD can show all of the engine information.
- It has a temperature probe to monitor system temperatures.
- In case of a DAU failure, there will be red Xs over the engine indications.
- MFD Limitations (PIM - section 09, page 9-64a):
 - The moving map display must not be used as the primary navigation instrument.
 - Use of the map page during IFR flight requires an IFR-approved GPS receiver and installation.
 - The Avidyne FlightMax EX5000 Multi-function Display Pilot's Guide must be available to the pilot during all flight operations.
 - Aircraft dispatch is prohibited when the MFD is inoperative.
 - Lightning overlay - N/A to our aircraft
- Autopilot:
 - S-Tec 55X:
 - The autopilot controls movement on two axes, longitudinal and lateral.
 - Attached to the cables and pulleys which move the control surfaces are 3 servos that act as actuators to increase or decrease tension based on commands sent from the autopilot.
 - Roll servo (located under the cabin floor)- Linked to turn coordinator.
 - Pitch servo (located under the baggage compartment floor)- Receives data from PFD, altimeter, VS, and pressure altitude transducer. Static pressure is received from the port located on both sides of the fuselage.
 - Trim servo (located behind the baggage compartment)- Electric trim must be on.
 - Control Wheel Steering (CWS)- Hold down the button on the yoke to disengage the servos and manipulate the controls, then release to lock servos in place.
 - When autopilot use is permitted:
 - The autopilot may be used at any time at or above 1,000 ft AGL and during the descent to the MDA or DA.

Aircraft Antennas

- Take the student to one of the Warrior Avidyne and show them which antenna is which.

Aircraft Preflight (IFR Emphasis)

- Instrument preflight check
 - Airspeed should read zero.
 - The altimeter reads within 75' of field elevation with the correct altimeter setting applied.
 - The attitude indicator becomes level within 5 minutes of vacuum pressure and does not bank more than 5 degrees on the ground in turns.
 - Turn coordinator is level with no flags while not turning, same direction while turning, and the ball swings to the opposite end.

- The vertical speed indicator reads zero; if not zero, use what's indicating as new zero.
- The compass should be full of fluid, free-floating, turn in the correct direction, and indicate a known heading.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 302.
- Assign Homework:
 - Emphasize to the student to begin reviewing now!

Completion Standards

At the completion of this ground lesson, the student should be familiar with weather, systems, and preflight operations for flight in IFR conditions.